# Exercise4

January 23, 2026

## 1 Exercise 4

[1]:
```
import numpy as np
# defining g
g = lambda w: np.dot(w.T,w) + 2

# function to generate random unit directions
def generate_random_directions(num_samples, D):
        # generating random directions
        directions = np.random.randn(num_samples, D)

        # normalizing to unit vectors
        directions = directions / np.linalg.norm(directions, axis=1)[:, np.
 ↪newaxis]

        return directions

# function returning the number of decreasing directions
def Number_of_decreasing_directions(g, w0, directions):
        decreasing_dirs = 0
        for i in range(directions.shape[0]):
                if g(w0 + 0.01 * directions[i, :]) < g(w0):
                        decreasing_dirs += 1
        return decreasing_dirs
```

[2]:
```
# function to compute and plot the number of decreasing directions vs dimension␣
 ↪N
def decreasing_directions_vs_dimension(g, P):
        # computer the number of decreasing directions for different dimensions␣
 ↪N
        decreasing_dirs = []
        for n in range(1, 26):
                # generate 100 random unit directions in R^n
                directions = generate_random_directions(P, n)

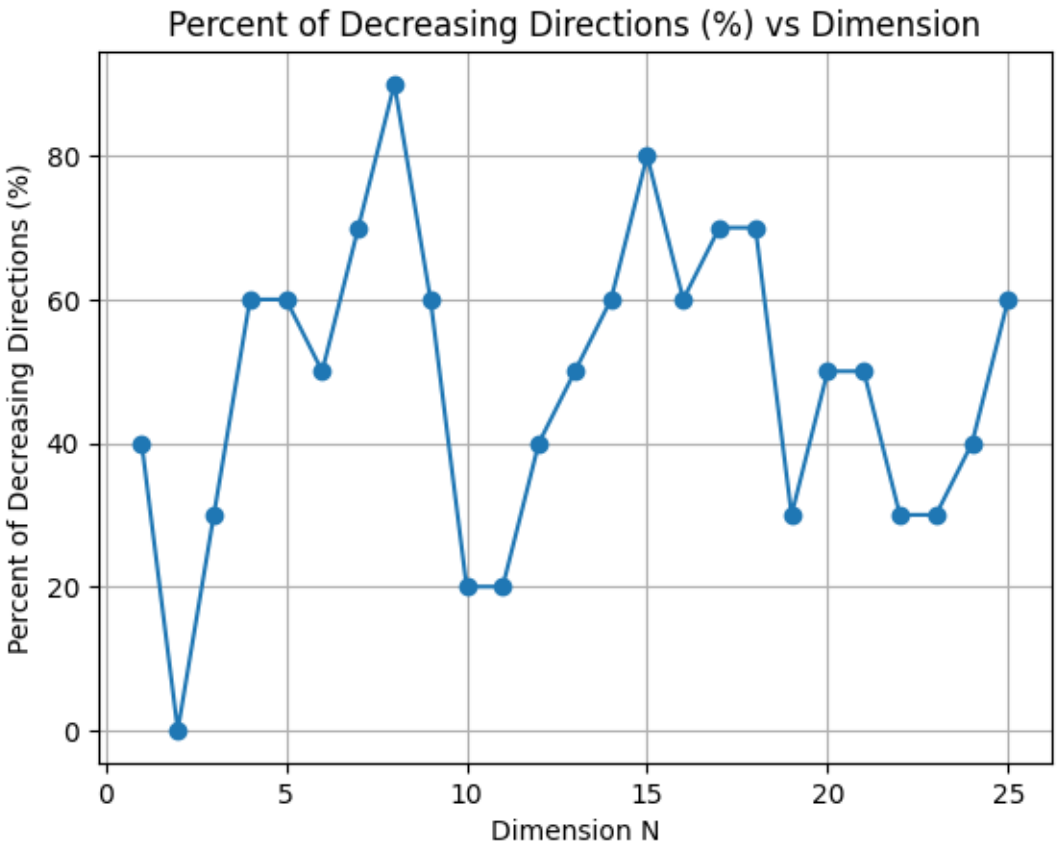
                # initialize w0
                w0 = np.zeros(n)
```

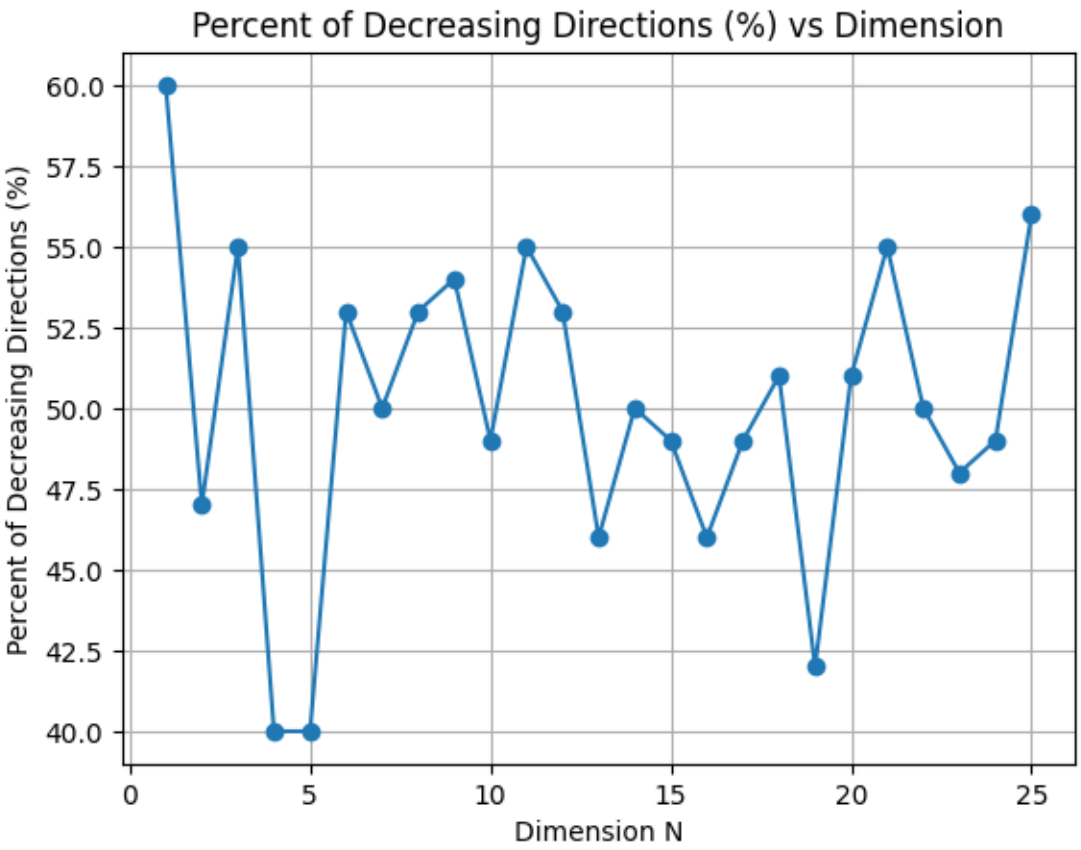
```
                w0[0] = 1

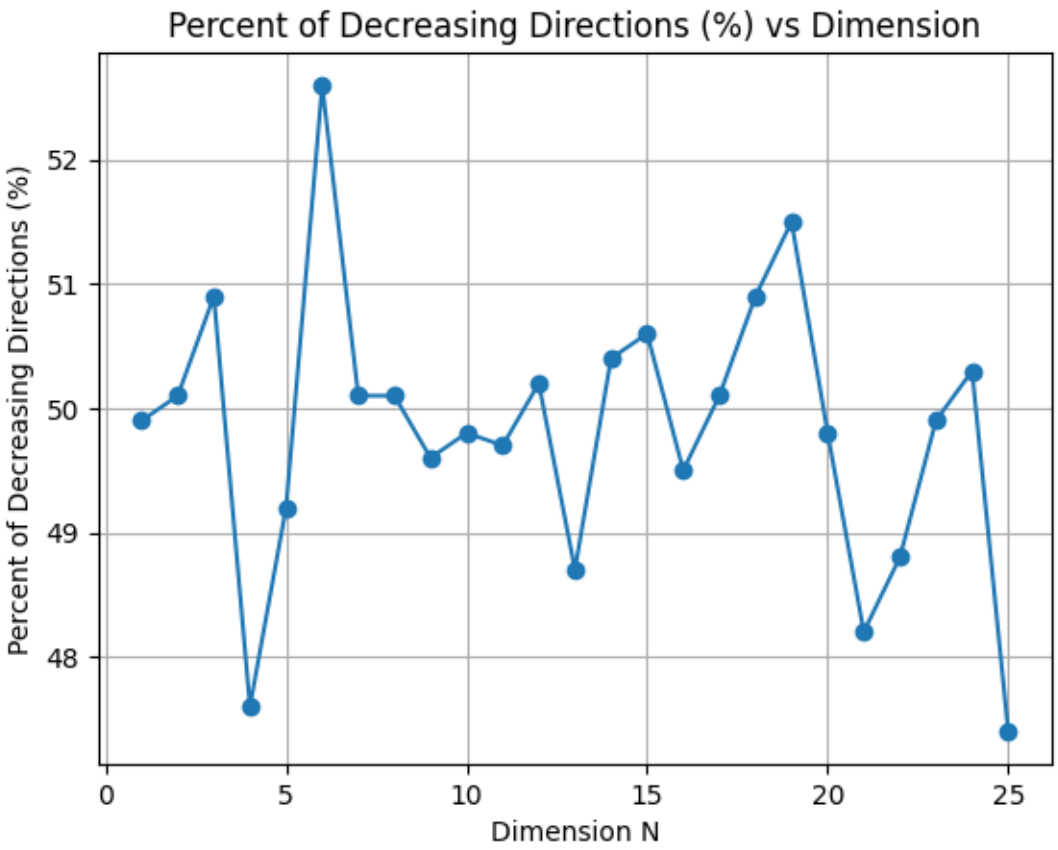
                # compute number of decreasing directions
                decreasing_dirs.append(Number_of_decreasing_directions(g, w0,␣
 ↪directions) * 100 / P)

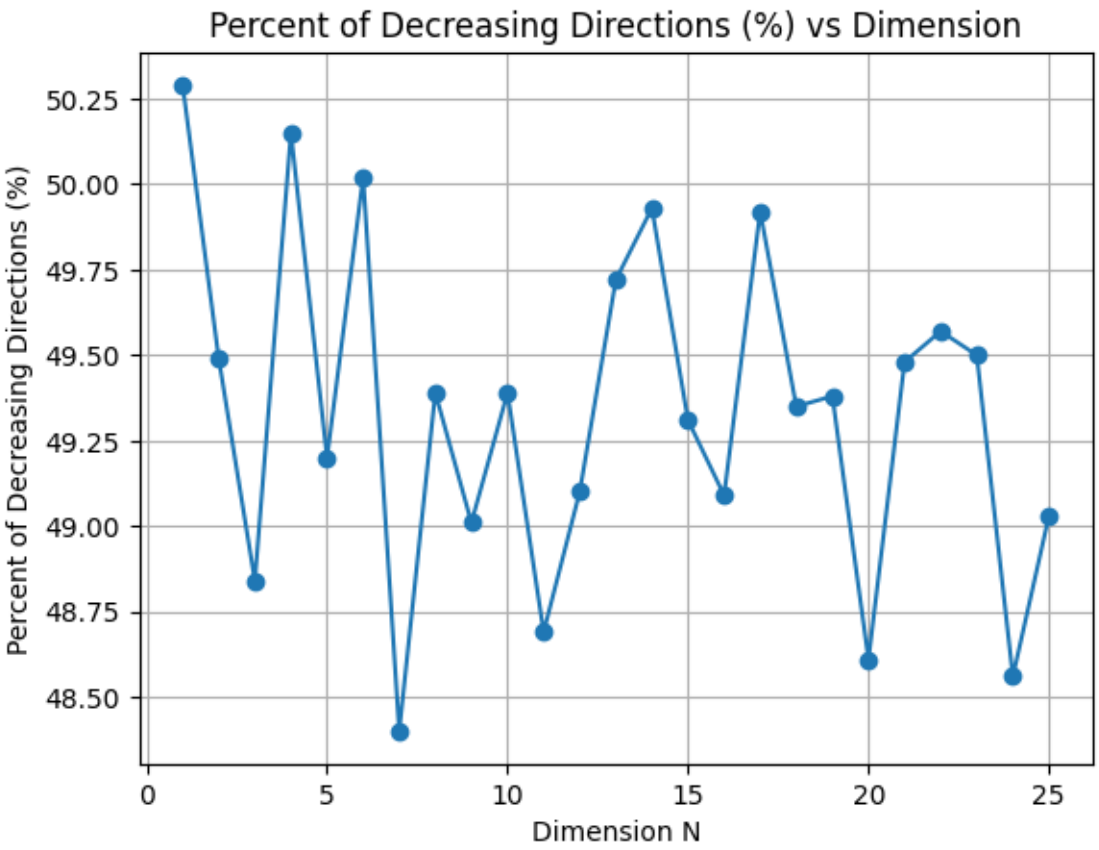
        # plot the results
        import matplotlib.pyplot as plt
        plt.plot(range(1, 26), decreasing_dirs, marker='o')
        plt.xlabel('Dimension N')
        plt.ylabel('Percent of Decreasing Directions (%)')
        plt.title('Percent of Decreasing Directions (%) vs Dimension')
        plt.grid()
        plt.show()

# call the function for different values of P
for P in [10, 100, 1000, 10000]:
        decreasing_directions_vs_dimension(g, P)
```




Percent of Decreasing Directions (%) vs Dimension
Percent of Decreasing Directions (%)
60.0
57.5
55.0
52.5
50.0
47.5
45.0
42.5
40.0
0
5
10
15
20
25
Dimension N


Percent of Decreasing Directions (%) vs Dimension
Percent of Decreasing Directions (%)
52
51
50
49
48
0
5
10
15
20
25
Dimension N




We observe that as the dimension N increases, the percentage of decreasing directions decreases significantly,indicating that in higher dimensions, it becomes increasingly difficult to find directions that lead to a decrease. The majority of decreasing directions are below 60% in all plots. Nevertheless we have some fluctuations.